COURSE UNIT 1-1

Air Pollution Knows No Borders

Air Quality in Taiwan

?

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Siaogang, Kaohsiung • scene photograph

Photo:吳初心

From Local Air to Shared Language

A Common Language for Air Pollution and Respiratory Health

1-1 **Taiwan air pollution status**

1-2 **Vietnam air pollution status**

1-3 **Major air pollutants and AQI interpretation**



Three Questions for Today

Describe a pattern, explain a case, and connect a measure to exposure.



Where and when?

Describe regional and seasonal differences.



Why does it change?

Identify sources and weather conditions.



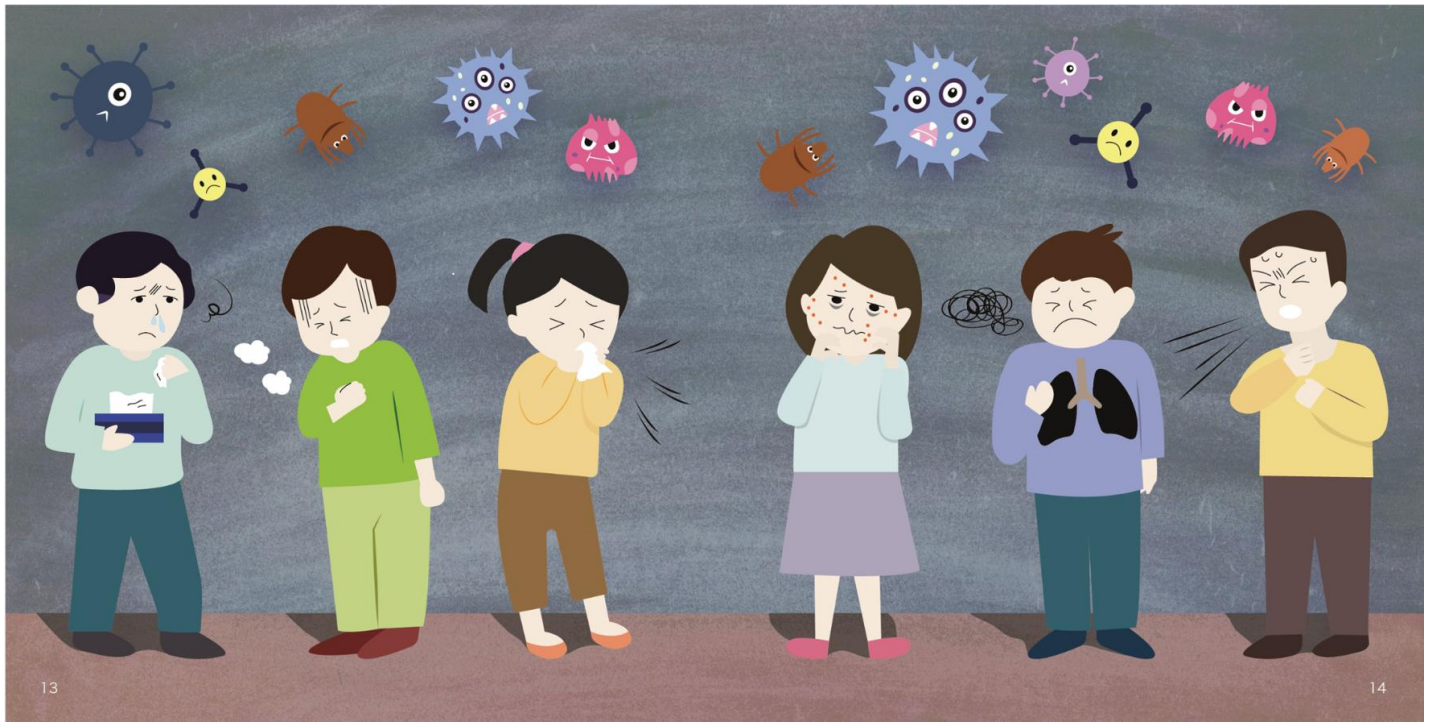
What can we do?

Link one control measure to reduced exposure.

Why Should We Care About Air Pollution?



**Air is invisible,
but its effects
happen every day.**



Air Pollution Around Us

What can you identify in this everyday scene?



Where Do Air Pollutants Come From?

Source categories and transport pathways help us move from observation to prevention.



Mixed urban sources

Natural releases also occur

Dust storms • volcanic activity • sea salt spray
Wildfires • rock weathering

01

Stationary sources

Factories • power plants
Boilers

02

Mobile sources

Cars • motorcycles
Trucks • ships

03

Fugitive / diffuse

Construction dust
Road dust • open burning

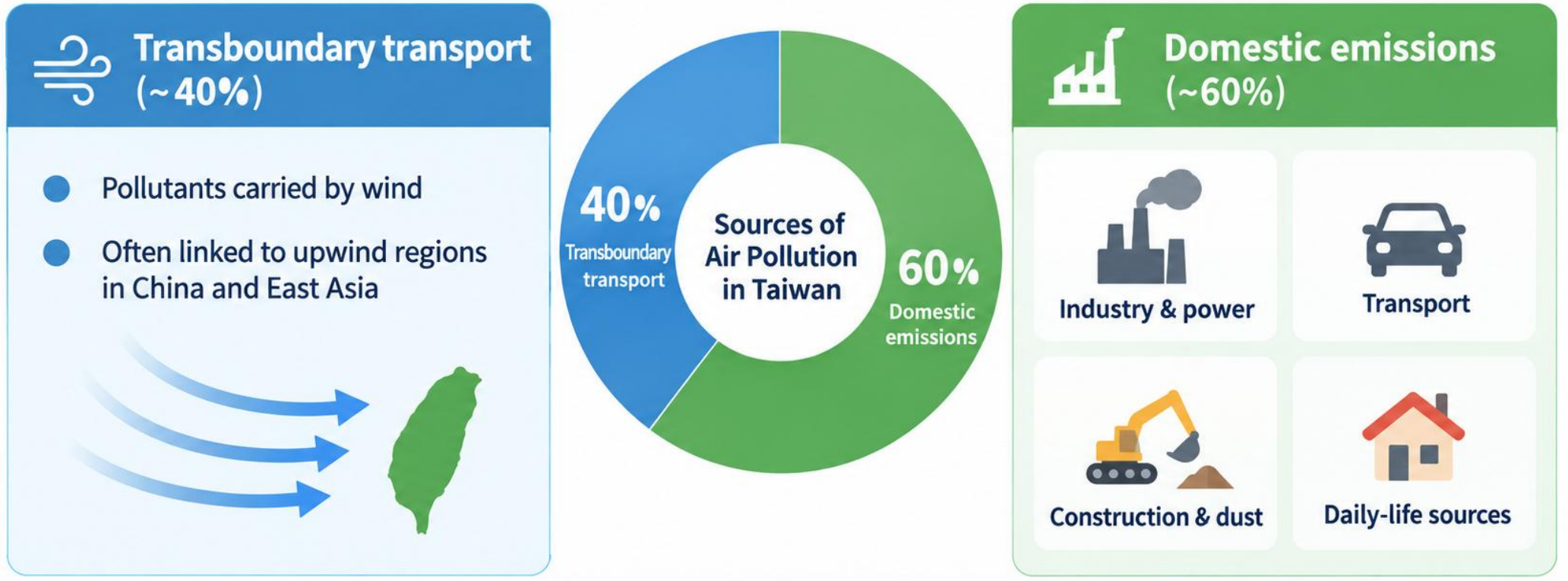
04

Transboundary transport

Regional pollutants carried by winds

Why Does Air Pollution Know No Borders?

Air moves, and pollution moves with terrain and weather.



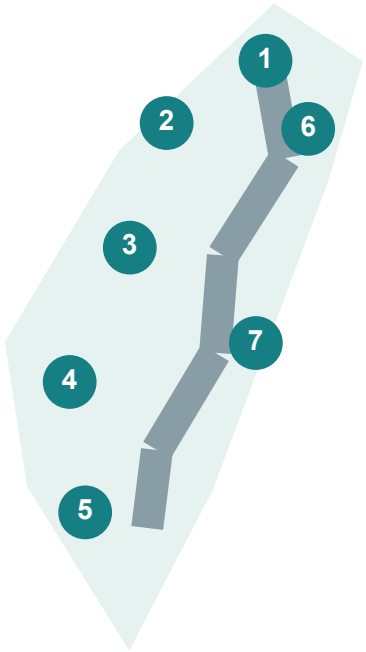
Taiwan's air pollution is influenced by both outside transport and local emissions.

Broad policy estimate—not fixed by day, pollutant, or region; the reference period is not shown in the supplied figure.

One Taiwan, Different Air

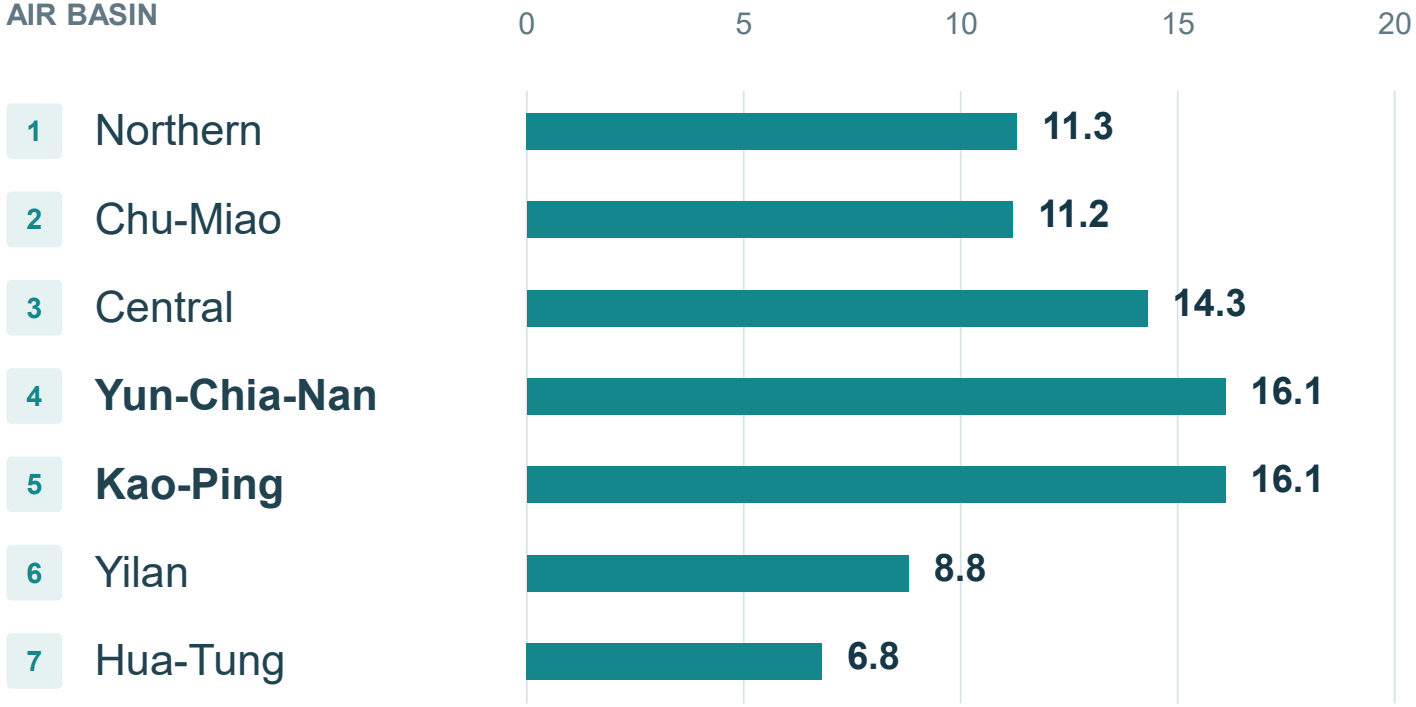
Same year. Same pollutant. Different air basins. | 2025 annual mean PM2.5, μg/m³

SEVEN AIR BASINS



Locator map; air-basin boundaries not drawn.

AIR BASIN



Annual concentrations—not AQI categories or individual exposure.

Taiwan’s Air Pollution Equation



Terrain

Central mountains influence airflow and pollutant dispersion.

Monsoon

Northeast monsoon can transport pollutants; accumulation depends on wind, terrain, and weather.

Local emissions

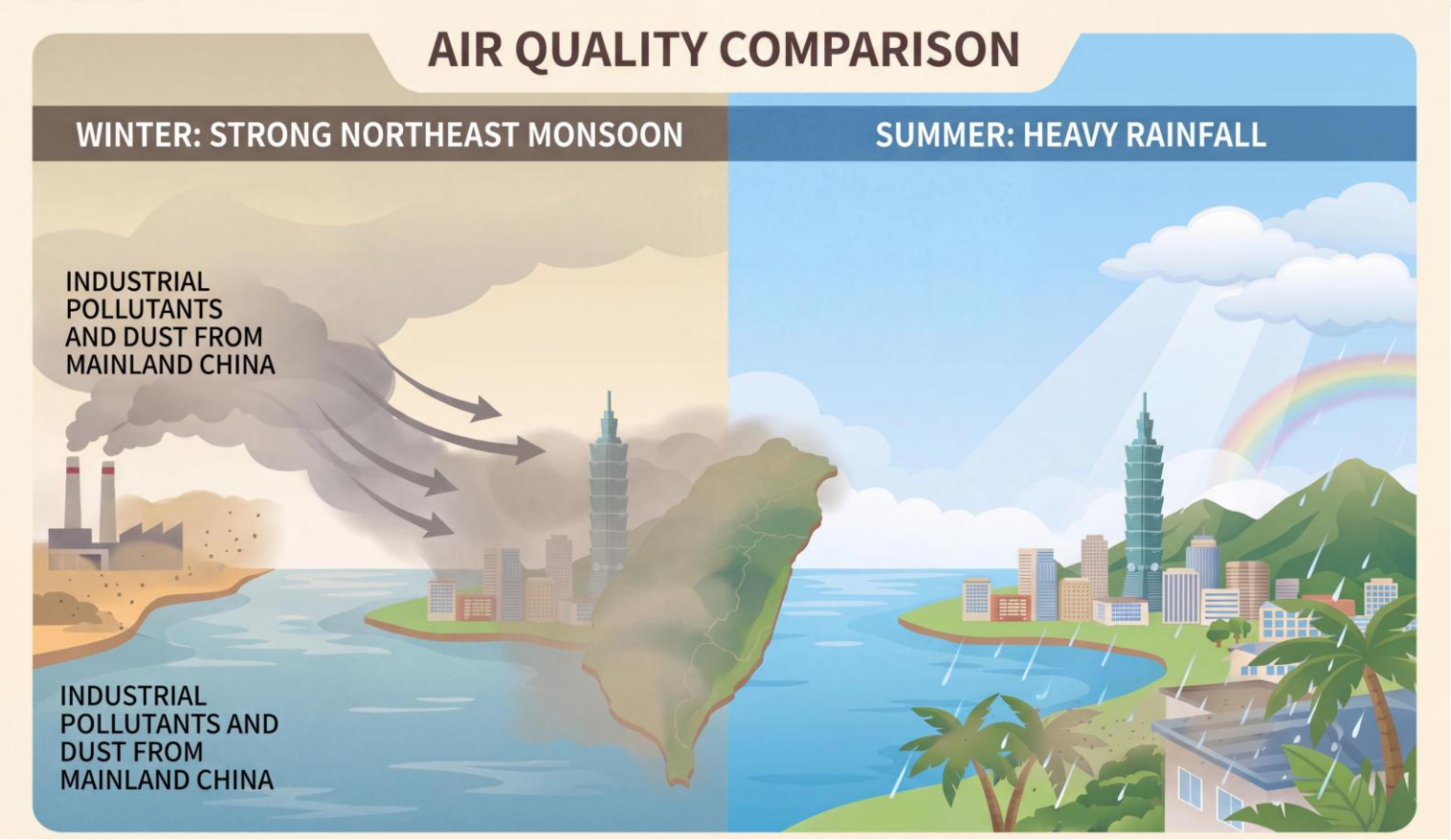
Industry, power generation, traffic, ports, and dust add local burden.

Transboundary transport

Air pollution can cross borders and seas.

Winter Monsoon vs. Summer Rainfall

How seasonal weather shapes air quality in Taiwan.



WINTER

Northeast monsoon

Transported pollution and dust

SUMMER

Heavy rainfall

Rainfall and pollutant removal

Seasonal illustration—not a forecast.

Same Place, Different Seasons

Qualitative pattern from MoENV's 2018–2022 monitoring summary—not 2025 monthly data.

WINTER



Higher PM2.5

Weaker mixing; regional transport

SPRING



Often higher

Transition weather; possible dust events

SUMMER



Often lower

More rain and stronger mixing

AUTUMN



Generally lower

Varies as the monsoon shifts



A different concern: sunlight can promote ground-level ozone—even when PM2.5 is lower.

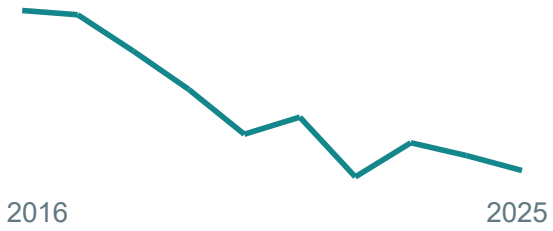
[2,3,6] Historical pattern • not a forecast

One Country, Three Ways to Read the Air

Two views of annual PM2.5, plus a short-term AQI snapshot. Each answers a different question.

01 LONG-TERM TREND

Change over years



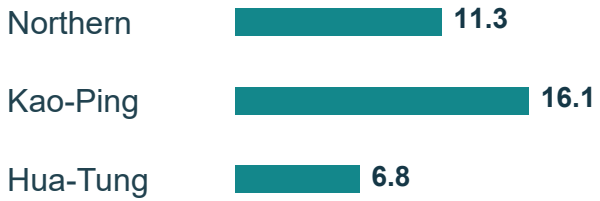
Annual PM2.5 • µg/m³

Same national series, several years.

EXPLORE SLIDE 13 →

02 REGIONAL PATTERN

Compare places



2025 • selected air basins

Annual PM2.5 • µg/m³

Same year and pollutant, different air basins.

REVISIT SLIDE 8 →

03 SHORT-TERM SNAPSHOT

Here and now

128 AQI
Unitless index

ILLUSTRATIVE ONLY • NOT LIVE

Current station AQI • no unit

Station + date/time. Updated hourly.

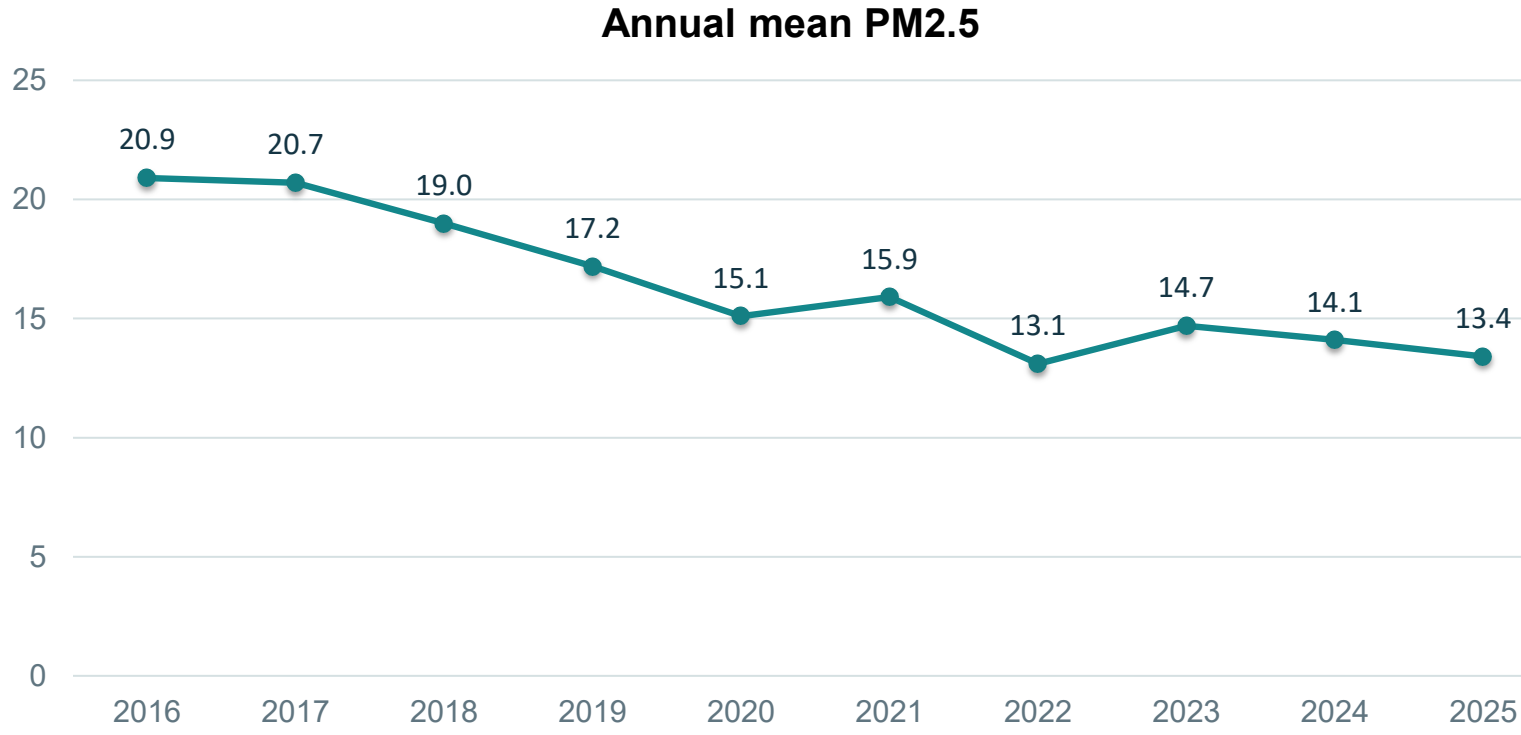
EXPLORE SLIDE 14 →

Annual PM2.5 concentration ≠ current AQI ≠ actual personal exposure.

[1,10,11] AQI card is illustrative

Taiwan’s PM2.5 Trend: What Has Changed?

Annual mean concentration • automated monitoring series, as published • µg/m³



OBSERVE FIRST

20.9

↓

13.4

2016 to 2025 • µg/m³

Lower overall.
Yearly variation.
Episodes remain.

[1] Table 3-18 • 2016–2025

Reading Taiwan's Real-Time AQI Map



TAIWAN AIR QUALITY MONITORING NETWORK

Ministry of Environment • official website

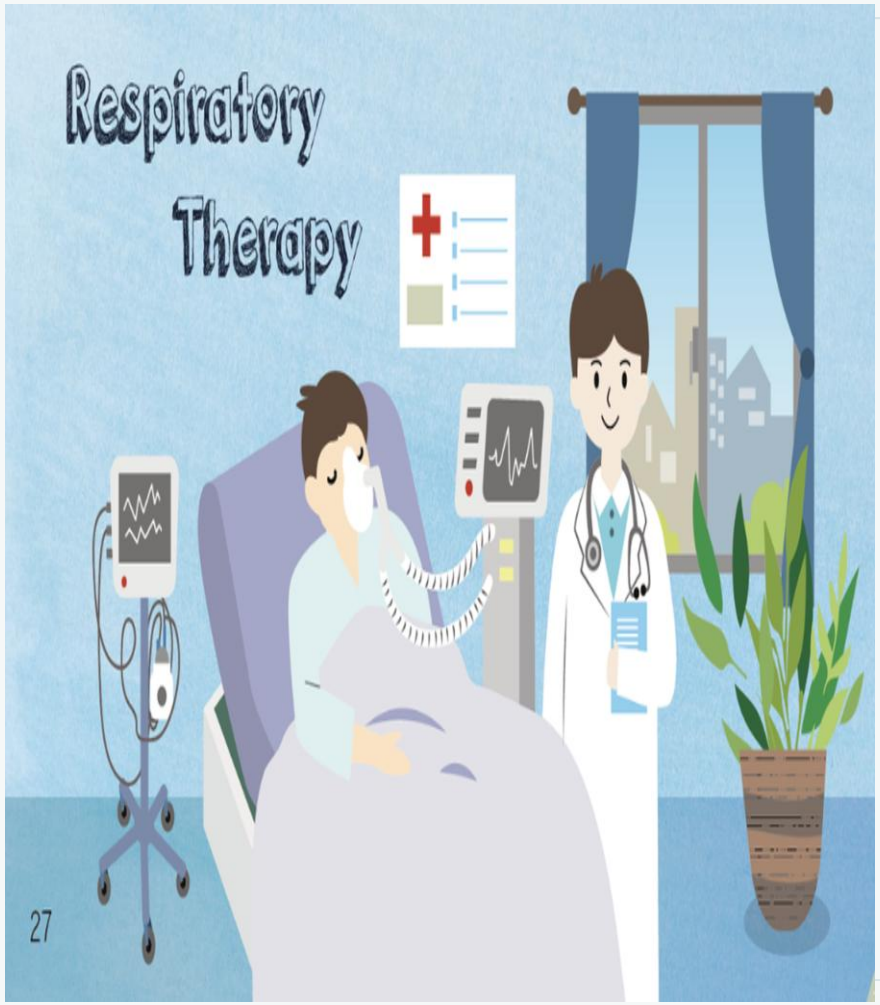
1. **Choose the variable**
   AQI is selected: a unitless index.
   PM2.5 and PM10 tabs show concentrations.
2. **Switch the map view**
   Color field: view the spatial pattern.
   Station dots: locate monitoring stations.
3. **Read the AQI color scale**
   Green = good; yellow = moderate.
   Orange = unhealthy for sensitive groups.
4. **Check the station and time**
   Open station details for the leading pollutant.
   Confirm the publication time on the website.

Open the official live map

AQI snapshot ≠ annual PM2.5

From Air Pollution to Health Burden

The lungs are a point of contact; health effects are not limited to the lungs.



Symptoms & function

Airway irritation • asthma worsening
Lower lung function

Disease & care

Respiratory and cardiovascular health risks

Long-term burden

Chronic disease • lung cancer
Premature mortality

[5,6] General evidence; no Taiwan count

How Taiwan Responds

Three complementary functions—not three interchangeable solutions.

01



Reduce emissions

Control pollution at the source.

Industry, vehicles, and dust controls.

02



Monitor and warn

Measure the air and communicate risk.

Monitoring, forecasts, and public alerts.

03



Reduce exposure

Use information to plan safer activity.

Adjust place, timing, and activity intensity.

[7,9] Supplied framework: 2024–2027

Control the Source, Not Only the Smoke

The three response functions become four practical policy steps.

1

Monitor

Air quality stations track pollutants and AQI.

→

2

Set targets

PM2.5 goals and standards guide policy.

→

3

Reduce emissions

Controls for industry, vehicles, vessels, dust, and VOCs.

→

4

Guide behavior

AQI alerts help people adjust outdoor activity.

Policy goal example: improve PM2.5 and reduce high-pollution days through coordinated source controls.

What Taiwan Teaches Us

A shared method for understanding air—not a single number to memorise.

WHERE + WHEN	Separate annual trends, regional averages, and short-term local conditions.
WHY	Consider sources, weather, terrain, and transported pollution together.
WHAT CAN HELP	Match each measure to the source and the exposure it aims to reduce.

Next: Vietnam • Two Cities, One Sky — Same questions. New context.

Sources and Teaching Notes

Reference slide—no narration needed. Source links, methods, photo provenance, and notes are retained.

1. **Official monitoring data**
   2025 annual report • trend and seven air-basin means
2. **Seasonality**
   2018–2022 historical pattern • qualitative summary
3. **Terrain, wind and regional patterns**
   MoENV regional and monsoon explanations
4. **Real-time AQI interface**
   MoENV Air Quality Monitoring Network • map and AQI guidance
5. **Particulate matter and health**
   US EPA • respiratory and cardiovascular effects
6. **Air quality and health**
   WHO • general evidence, not Taiwan case counts
7. **Taiwan policy framework**
   Air Pollution Control Plan 2024–2027
8. **Source-control examples**
   MoENV regulations • industry, transport, and dust controls

Photos: supplied Da-Wei Wu lecture; dates not recorded. Maps: Natural Earth. Diagrams: teaching schematics.

For today’s decisions, check current official monitoring—not an archived slide.

From Air-Quality Knowledge to Environmental Education



Picture book excerpt

How the story supports this unit

Images help learners connect sources, exposure, health effects, and protective actions.



Everyday sources



Health effects



Indoor choices



Action & education

Two Cities, One Sky

Taiwan → Vietnam

UNIT 1-1

Taiwan

THIS LESSON

UNIT 1-2

Vietnam

COMING NEXT



Thank You for Your Attention

Air Pollution and Respiratory Health

Air Quality in Taiwan

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